Week 6 - Graded Assignment 6

The due date for submitting this assignment has passed.
Due on 2026-03-25, 23:59 IST.
You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-03-25, 07:24 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

Multiple Select Questions (MSQ):

1 point
A function $T: V \rightarrow W$T : V → W between two vector spaces VV and WW is said to be a linear transformation if the following conditions hold:

- Condition 1:Condition 1: $T(v_1 + v_2) = T(v_1) + T(v_2)$T(v1 + v2) = T(v1) + T(v2) for all $v_1, v_2 \in V$v1, v2 ∈ V.
- Condition 2:Condition 2: $T(cv) = cT(v)$T(cv) = cT(v) for all $v \in V$v ∈ V and $c \in R$c ∈ R.

Consider the following function:

$$T: R^2 \rightarrow R \quad \text{T : R2 → R}$$
$$T(x, y) = \begin{cases} 3x & \text{if } y = 0 \\ 4y & \text{if } y \neq 0 \end{cases} \quad \text{T(x, y) = \{3x if y = 0 4y if y ≠ 0}$$

Which of the following statements is true?
- [ ] Condition 1 holds.
- [x] Condition 1 does not hold.
- [x] Condition 2 holds.
- [ ] Condition 2 does not hold.

Yes, the answer is correct.
Score: 1
Accepted Answers:
Condition 1 does not hold.
Condition 2 holds.

1 point
Suppose the matrix representation of a linear transformation $T: R^3 \rightarrow R^3$T : R3 → R3 with respect to ordered bases $\beta = \{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$β = {(1, 0, 1), (0, 1, 0), (0, 0, 1)} for the domain and $\gamma = \{(1, 0, 0), (0, 1, 0), (1, 0, 1)\}$γ = {(1, 0, 0), (0, 1, 0), (1, 0, 1)} for the range, is $I_{3\times 3}$I3×3, i.e., the identity matrix of order 3. Let AA denote the matrix representation of the linear transformation TT with respect to the standard ordered basis of R^3R3 for both domain and range. Which of the following are true?

- [ ] $A = I_{3\times 3}$A = I3×3 i.e., identity matrix of order 3.
- [ ] AA is a singular matrix

$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$A = $\begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 \end{bmatrix}$

- [x] $det(A)$ = 1det(A) = 1.
- [ ] $det(A) = -1$ = −1det(A) = −1.
- [x] $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$A = $\begin{bmatrix} 0 & 0 & -1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$

Yes, the answer is correct.
Score: 1
Accepted Answers:
$det(A)$ = 1det(A) = 1.
$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$A = $\begin{bmatrix} 0 & 0 & -1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$

1 point
Match the linear transformations and sets of vectors in column A with the images of those sets under the linear transformation in column B and the geometric representations of both sets in column C.

Choose the correct option from the following.

- [ ] i → → d → → 2, ii → → b → → 4.
- [x] i → → b → → 4, ii → → d → → 2.
- [x] iii → → a → → 1, iv → → c → → 3
- [ ] iii → → a → → 3, iv → → c → → 1

No, the answer is incorrect.
Score: 0
Accepted Answers:
i → → b → → 4, ii → → d → → 2.
iii → → a → → 3, iv → → c → → 1

1 point
Consider a linear transformation $S: M_2(R) \rightarrow M_2(R)$S : M2(R) → M2(R) such that $S(A) = A^T$ S(A) = AT . Let BB be the matrix representation of SS with respect to the ordered bases:

$$\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\}\{[1000], [0010], [0100], [0001]\}$$

of $M_2(R)$M2(R). Choose the set of correct options:

- [ ] The order of the matrix BB is $2 \times 2$2 × 2.

- [x] The order of the matrix BB is $4 \times 44 \times 4$.
- [x] The dimension of the row space of the the matrix BB is 4.
- [ ] The dimension of the column space of the matrix BB is 3.
- [ ] The nullity of the the matrix BB is 1.
- [x] The rank of the the matrix BB is 4.
- [x] SS is surjective.

Yes, the answer is correct.
Score: 1
Accepted Answers:
The order of the matrix BB is $4 \times 44 \times 4$.
The dimension of the row space of the the matrix BB is 4.
The rank of the the matrix BB is 4.
SS is surjective.

1 point

Let $T: R^n \rightarrow R^m$ T : Rn → Rm be a linear transformation and tet $\{v_1, v_2, \ldots, v_n\}$ {v1, v2, … , vn} be a basis of R^n Rn. Suppose $\{T(v_1), T(v_2) \ldots, T(v_n)\}$ {T (v1), T (v2) … , T (vn)} is a spanning set for R^m Rm. Choose all the correct statements.

- [x] TT is a surjective linear transformation.
- [ ] TT is a linear isomorphism.
- [x] $m \leq n$m ≤ n.
- [x] For any basis $\{w_1, w_2, \ldots, w_n\}$ {w1, w2, … , wn} of R^n Rn, the set $\{T(w_1), T(w_2), \ldots, T(w_n)\}$ {T (w1), T (w2), … , T (wn)} is a spanning set for R^m Rm.
- [ ] The matrix of the linear transformation TT has rank equal to nn.

Yes, the answer is correct.
Score: 1
Accepted Answers:
TT is a surjective linear transformation.
$m \leq n$m ≤ n.
For any basis $\{w_1, w_2, \ldots, w_n\}$ {w1, w2, … , wn} of R^n Rn, the set $\{T(w_1), T(w_2), \ldots, T(w_n)\}$ {T (w1), T (w2), … , T (wn)} is a spanning set for R^m Rm.

Comprehension Type Question:

Suppose a bread-making machine B makes 6 breads from 2 eggs, 3 (in hundreds) grams of wheat, and 1 (in hundred) grams of sugar. B also makes 8 breads from 3 eggs, 4 (in hundreds) grams of wheat, and 2 (in hundreds) grams of sugar, and 10 breads from 5 eggs, 5 (in hundreds) grams of wheat, and 3 (in hundreds) grams of sugar. Suppose the production of breads is a linear function of the amount of eggs, wheat (in hundreds), and sugar (in hundreds) used as raw ingredients. Based on the above data answer the following questions. Suppose xx eggs, yy (in hundreds) grams of wheat, and zz (in hundreds)

grams of sugar are used as the raw materials to produce $ax + by + cz$ax + by + cz number of breads. We can express this as follows:

$$T: R^3 \rightarrow R\text{T} : \text{R3} \rightarrow \text{R}$$

$$T(x, y, z) = ax + by + cz\text{T}(\text{x}, \text{y}, \text{z}) = \text{ax} + \text{by} + \text{cz}$$

where the co-ordinates in R^3 R3 denote the number of eggs, amount (in grams) of wheat (in hundreds), and amount (in grams) of sugar (in hundreds). Observe that TT is a linear transformation.

Use the above information for Question 6, 7 and 8.

1 point

Choose the correct set of options from the the following. (MSQ)

- [ ] $Nullity(T)$ = 1Nullity(T) = 1
- [x] $Rank(T)$ = 1Rank(T) = 1
- [x] $Nullity(T)$ = 2Nullity(T) = 2
- [ ] $Rank(T)$ = 2Rank(T) = 2
- [ ] $Nullity(T)$ = 3Nullity(T) = 3
- [ ] $Rank(T)$ = 3Rank(T) = 3
- [ ] TT is neither one to one nor onto.
- [ ] TT is one to one but not onto.
- [x] TT is onto but not one to one.
- [ ] TT is an isomorphism.

Yes, the answer is correct.

Score: 1

Accepted Answers:

$Rank(T)$ = 1Rank(T) = 1

$Nullity(T)$ = 2Nullity(T) = 2

TT is onto but not one to one.

1 point

Choose the set of correct statements. (MSQ)

- [ ] If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then 9 breads are produced.
- [x] If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then no bread is produced.
- [x] If only 3 (in hundreds) grams of wheat is used, then 6 breads are produced.
- [ ] If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then no bread is produced.
- [x] If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then 6 breads are produced.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then no bread is produced.

If only 3 (in hundreds) grams of wheat is used, then 6 breads are produced.
If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then 6 breads are produced.
How many breads are produced by the machine from 6 eggs, 10 (in hundreds) grams of wheat, and 5 (in hundreds) grams of sugar? (NAT)

20

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 20
1 point

Let TT be a linear transformation from R^{10} R10 to R^{12} R12. Suppose a basis for the null space of TT has 33 vectors. How many linearly independent vectors are needed to form a basis for the range of TT?

7

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 7
1 point

Let MM be the set of all skew-symmetric matrices of order 7. Then MM forms a vector space under matrix addition and scalar multiplication. Let TT be a linear transformation from MM to RR defined by $T(A)$ = 6 $tr(A)$T(A) = 6 tr(A), where $tr(A)$ = tr(A) =trace of AA. What is the nullity of TT?

21

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 21.0
1 point

1 point
Let $T: R^3 \longrightarrow R^2$ T : R3 ⟶ R2 be defined by $T(x, y, z)$ = (10x + 9y, 2z)T(x, y, z) = (10x + 9y, 2z). Choose the correct options about TT.

☐ The matrix of TT is $\begin{bmatrix} 10 & 0 \\ 9 & 0 \\ 0 & 2 \end{bmatrix}$ $\begin{bmatrix} & \\ 1090002 & \end{bmatrix}$.

☑ The matrix of TT is $\begin{bmatrix} 1090 \\ 0\ 02 \end{bmatrix}$[1009002].

☑ A basis for the null space of TT is $\{(1, -\frac{10}{9}, 0)\}\{(1, -910, 0)\}$.

☐ A basis for the null space of TT is $\{(1, -\frac{10}{9}, 0), (0, 0, 2)\}\{(1, -910, 0), (0, 0, 2)\}$.

☐ A basis for the range of TT is $\{(10, 10)\}\{(10, 10)\}$.

☑

A basis for the range of TT is $\{(10, 9), (0, 2)\}\{(10, 9), (0, 2)\}$.

Yes, the answer is correct.
Score: 1
Accepted Answers:
The matrix of TT is $[\begin{matrix}1090\\0\ 02\end{matrix}][1009002]$.

A basis for the null space of TT is $\{(1, -\frac{10}{9}, 0)\}\{(1, -910, 0)\}$.